

Energy from Biomass: The conversion technologies

- The use of biomass to provide energy has been fundamental to the development of civilisation.
- In recent times pressures on the global environment have led to calls for an increased use of renewable energy sources, in lieu of fossil fuels.
- Biomass is one potential source of renewable energy and the conversion of plant material into a suitable form of energy, usually electricity or as a fuel for an internal combustion engine, can be achieved using a number of different routes, each with specific pros and cons.
- As with any energy resource there are limitations on the use and applicability of biomass and it must compete not only with fossil fuels but with other renewable energy sources such as wind, solar and wave power.
- Biomass can be converted into three main products:
 - ❖ two related to energy—power/heat generation and transportation fuels
 - ❖ and one as a chemical feedstock (summarized in **Figure-1**)

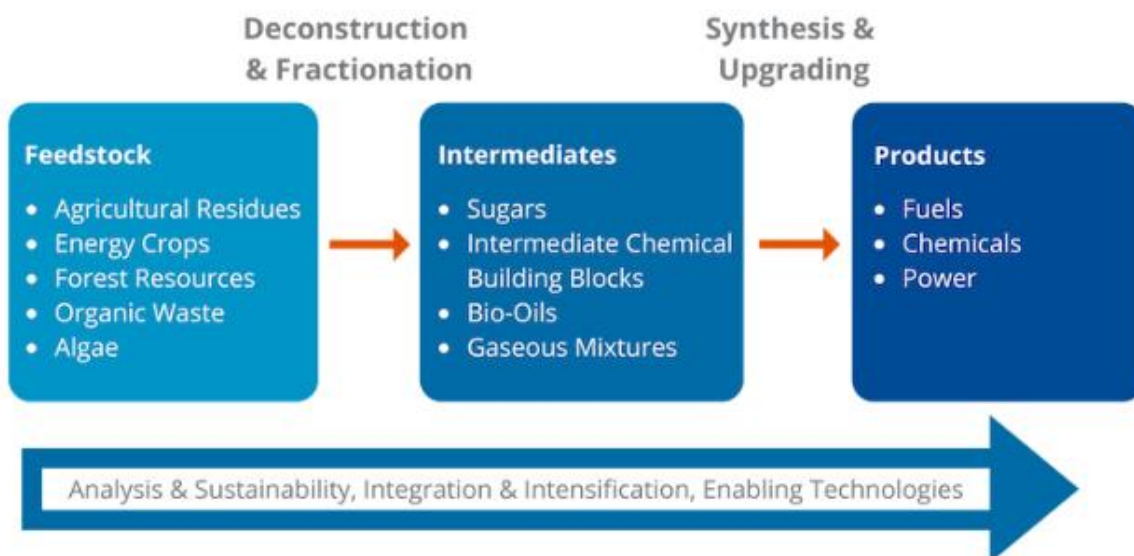


Figure-1



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➤ Bio-energy conversion: process options

- Biomass can be converted into useful forms of energy using a number of different processes.
- Factors that influence the choice of conversion process are:
 - ❖ the type and quantity of biomass feedstock;
 - ❖ the desired form of the energy, i.e. end-use requirements;
 - ❖ environmental standards;
 - ❖ economic conditions;
 - ❖ and project specific factors.

NOTE: In many situations it is the form in which the energy is required that determines the process route, followed by the available types and quantities of biomass.

➤ BIOMASS CONVERSION TECHNOLOGIES

- Conversion of biomass to energy is undertaken using two main process technologies:
 - ❖ thermo-chemical and
 - ❖ bio-chemical/biological
 - ❖ mechanical extraction is the third technology for producing energy from biomass

NOTE: Currently the cost of bio-diesel compared with fossil fuel makes the technology uncompetitive but increasing environmental pressures to improve air quality, especially in cities, may change this situation in near future.

- Within thermo-chemical conversion four process options are available: combustion, pyrolysis, gasification and liquefaction.

- Bio-chemical conversion encompasses two process options: digestion (production of biogas, a mixture of mainly methane and carbon dioxide) and fermentation (production of ethanol).
- A distinction can be made between the energy carriers produced from biomass by their ability to provide heat, electricity and engine fuels.
- A useful means of comparing biomass and fossil fuels is in terms of their O:C and H:C ratios, known as a Van Krevlen diagram, shown in **Figure-2**.
- The lower the respective ratios the greater the energy content of the material.

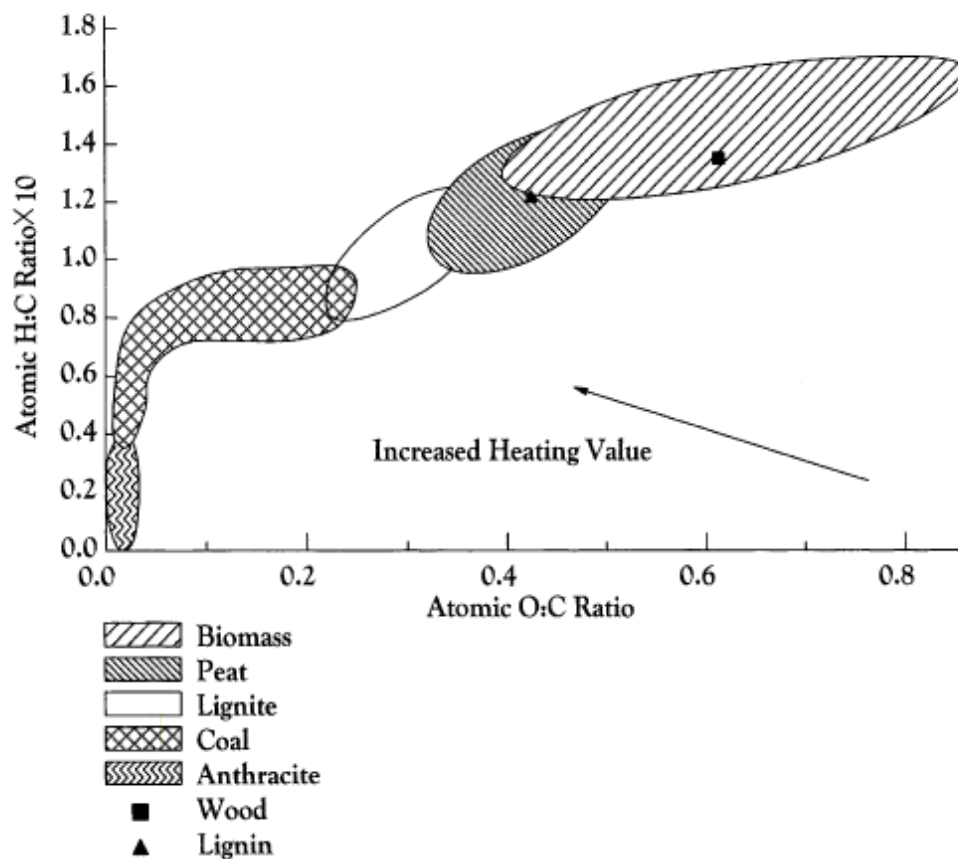
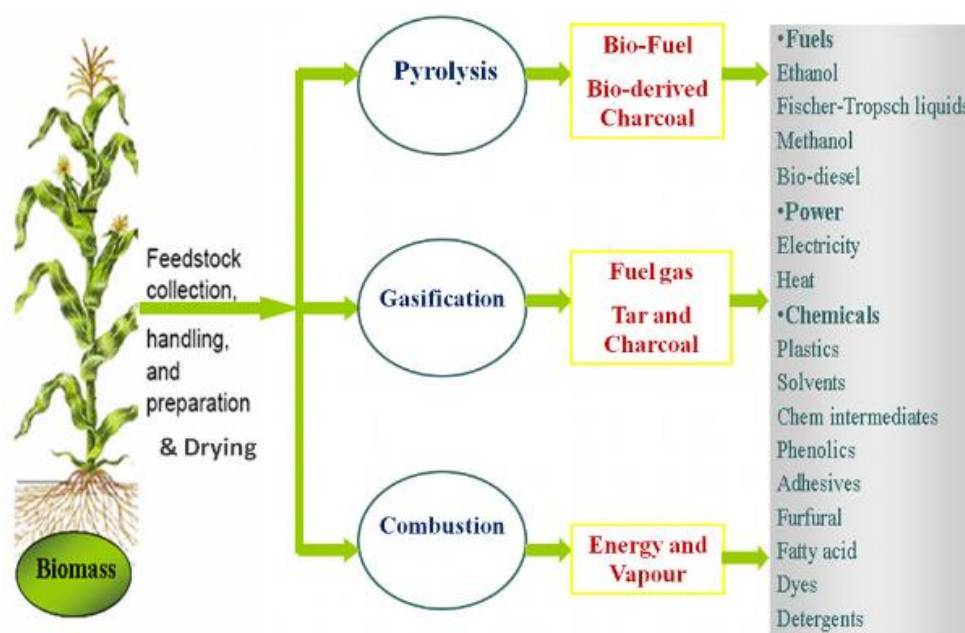


Figure-2: Van Krevelen diagram for various solid fuels.

➤ **Thermo-chemical conversion**

- Three main processes are used for the thermo-chemical conversion of biomass, together with two lesser-used options.
- The main processes, the intermediate energy carriers and the final energy products resulting from thermo-chemical conversion are summarized in **Figure-3** and flowchart shown in **Figure-4**.



Thermochemical Conversion Technologies for Biomass

Figure-3

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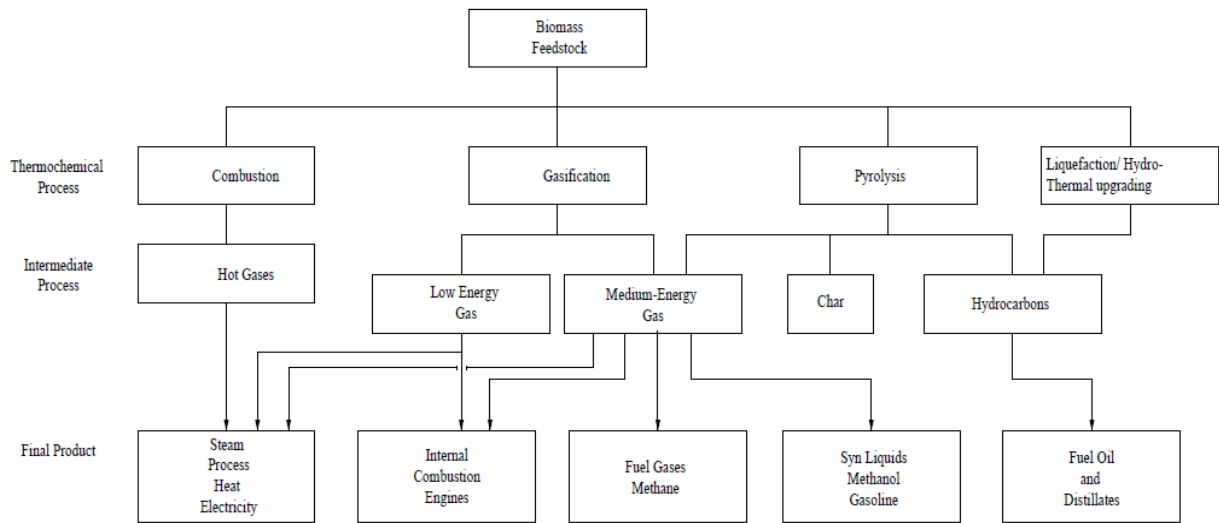


Figure-4

• **Combustion**

- ❖ The burning of biomass in air, i.e. combustion, is used over a wide range of outputs to convert the chemical energy stored in biomass into heat, mechanical power, or electricity (**Figure-5**).
- ❖ Various items of process equipment, e.g. stoves, furnaces, boilers, steam turbines, turbo-generators, etc. Combustion of biomass produces hot gases at temperatures around 800–1000 °C.
- ❖ It is possible to burn any type of biomass but in practice combustion is feasible only for biomass with a moisture content <50%, unless the biomass is pre-dried.
- ❖ High moisture content biomass is better suited to biological conversion processes.
- ❖ Net bio-energy conversion efficiencies for biomass combustion power plants range from 20% to 40%.
- ❖

Combustion Reaction

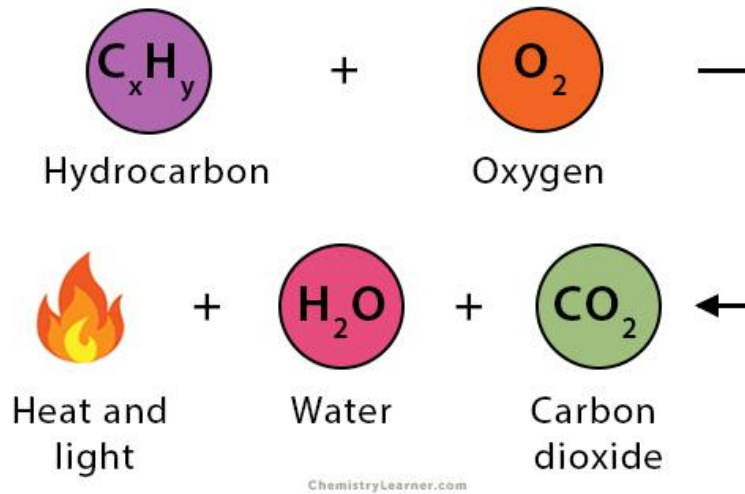


Figure-5

• Gasification

- ❖ Gasification is the conversion of biomass into a combustible gas mixture by the partial oxidation of biomass at high temperatures (Figure-6 & Figure-7)
- ❖ Typically in the range 800–900 °C. The low calorific value (CV) gas produced (about 4–6 MJ/N m³) can be burnt directly or used as a fuel for gas engines and gas turbines.
- ❖ The product gas can be used as a feedstock (syngas) in the production of chemicals (e.g. methanol).
- ❖ One promising concept is the **biomass integrated gasification/combined cycle (BIG/CC)**, where gas turbines convert the gaseous fuel to **electricity** with a high overall conversion efficiency.

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- ❖ The production of syngas from biomass allows the production of methanol and hydrogen, each of which may have a future as fuels for transportation.

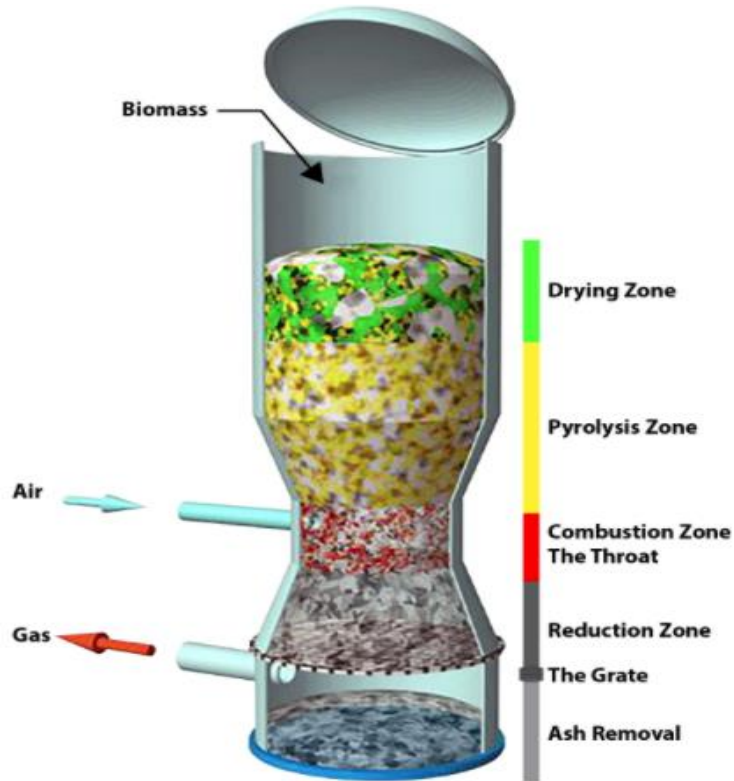


Figure-6

4 Processes in Gasification

not necessarily in order

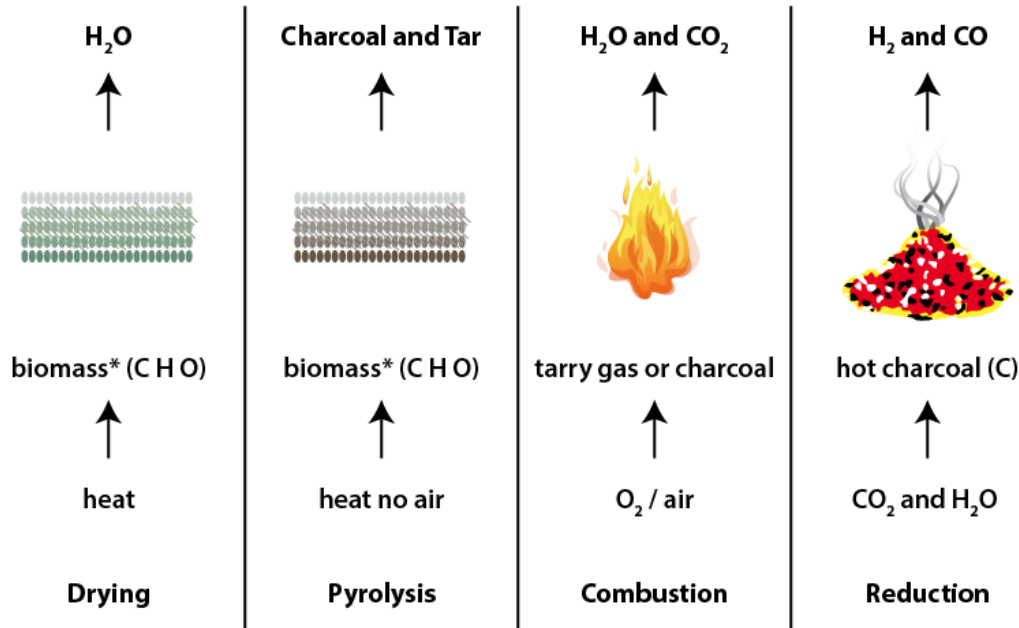


Figure-7

- **Pyrolysis**

- ❖ Pyrolysis is the conversion of biomass to **liquid** (termed bio-oil or bio-crude), **solid and gaseous fractions**, by heating the biomass in the **absence of air to around 500 °C** (**Figure-8 & Figure-9**).
- ❖ **Figure-8** depicts the range and possible yields of pyrolysis energy products.
- ❖ Pyrolysis can be used to produce predominantly **bio-oil if flash pyrolysis is used**, enabling the conversion of biomass to bio-crude with an efficiency of up to 80%.
- ❖ The bio-oil can be used in engines and turbines and its use as a feedstock for refineries is also being considered.

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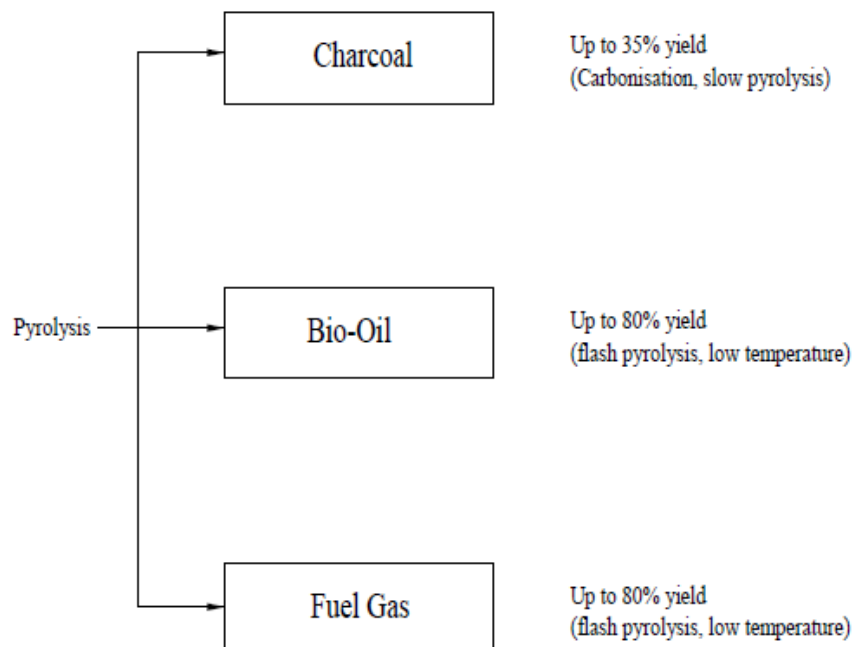


Figure-8: Energy products from pyrolysis.

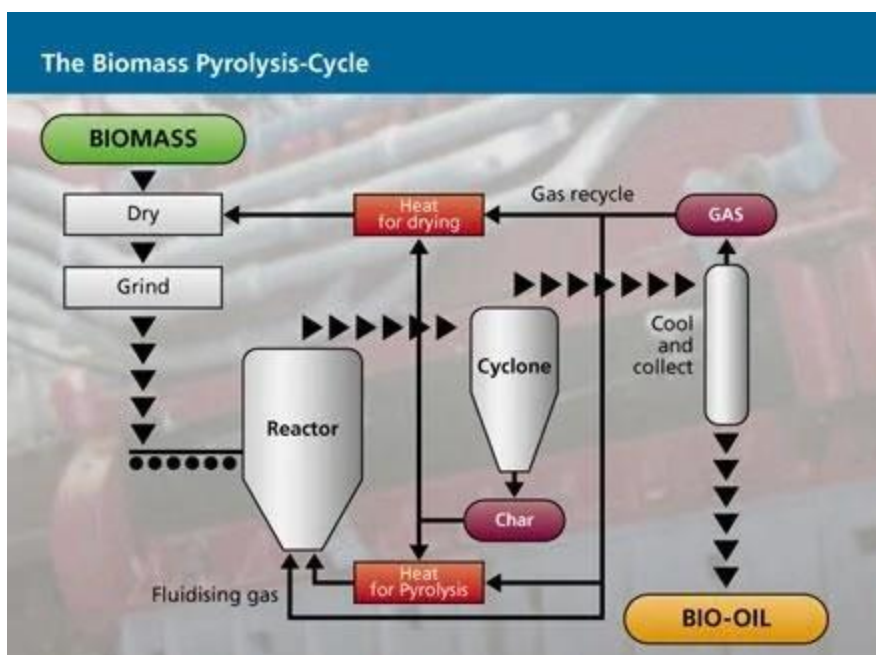


Figure-9

- ❖ Problems with the conversion process and subsequent use of the oil, such as its poor thermal stability and its corrosivity, still need to be overcome.
- ❖ Upgrading bio-oils by lowering the oxygen content and removing alkalis by means of hydrogenation and catalytic cracking of the oil may be required for certain applications.
- ❖ Possible treatment and upgrading options for bio-oil are shown in

Figure-10.

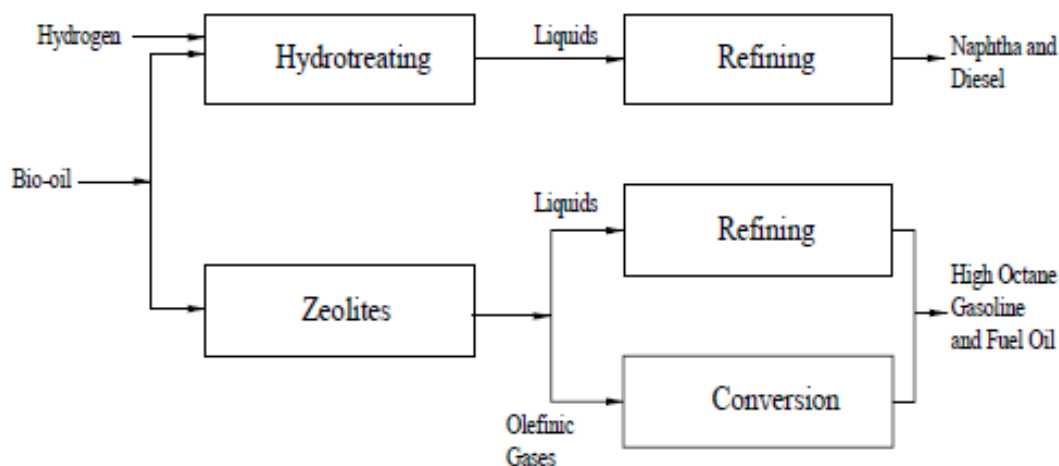


Figure-10: Treatment and upgrading options for bio-oil.

NOTE: Other processes that produce bio-oils are hydro thermal upgrading (HTU) and liquefaction. HTU converts biomass in a wet environment at high pressure to partly oxygenated hydrocarbons: the process is believed to be almost at the pilot stage. The interest in liquefaction is low because the reactors and fuel-feeding systems are more complex and more expensive than pyrolysis processes. Liquefaction is the conversion of biomass into a stable liquid hydrocarbon using low temperatures and high hydrogen pressures. A generalised conceptual flow sheet for liquefaction is shown in **Figures-11 & 12.**

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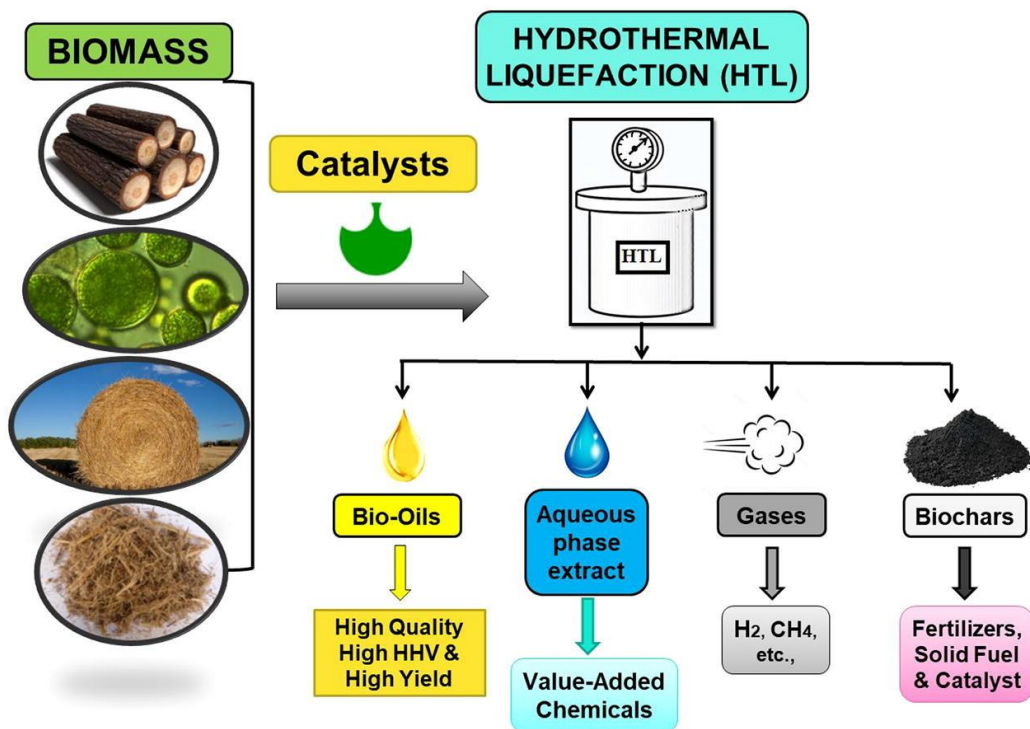


Figure-11

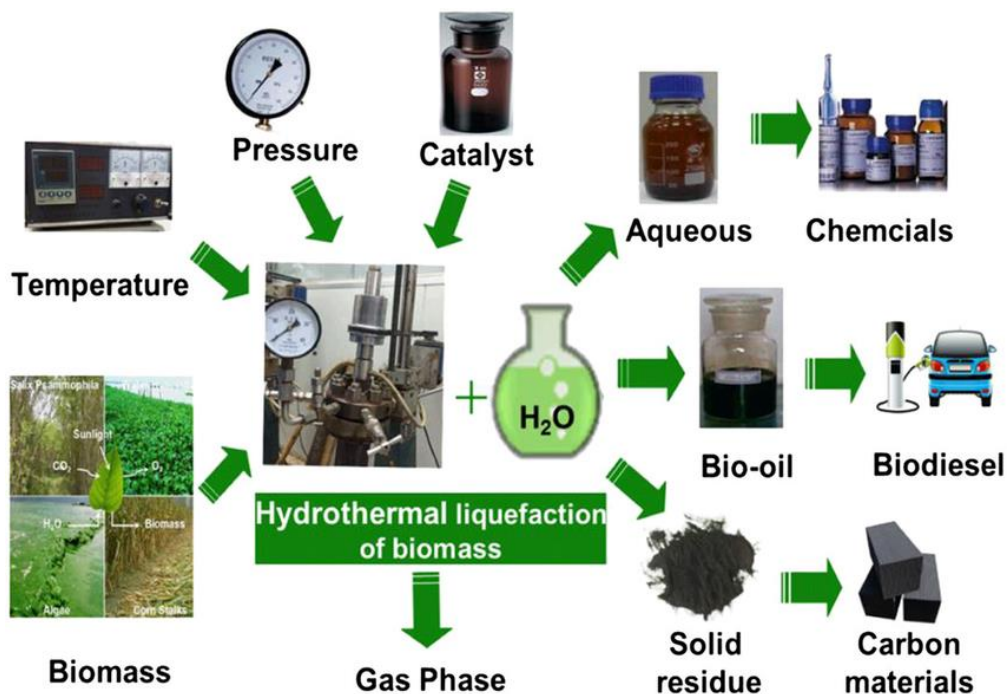


Figure-12

➤ **Bio-chemical conversion**

• **Fermentation**

- ❖ Fermentation is used commercially on a large scale in various countries to produce ethanol from sugar crops (e.g. sugar cane, sugar beet) and starch crops (e.g. maize, wheat) (**Figure-13**).
- ❖ The biomass is ground down and the starch converted by enzymes to sugars, with yeast then converting the sugars to ethanol.
- ❖ Purification of ethanol by distillation is an energy-intensive step, with about 450 l of ethanol being produced per ton of dry corn.
- ❖ The solid residue from the fermentation process can be used as cattle-feed and in the case of sugar cane, the bagasse can be used as a fuel for boilers or for subsequent gasification.
- ❖ The conversion of lignocellulosic biomass (such as wood and grasses) is more complex, due to the presence of longer-chain polysaccharide molecules and requires acid or enzymatic hydrolysis before the resulting sugars can be fermented to ethanol.
- ❖ Such hydrolysis techniques are currently at the pre-pilot stage.

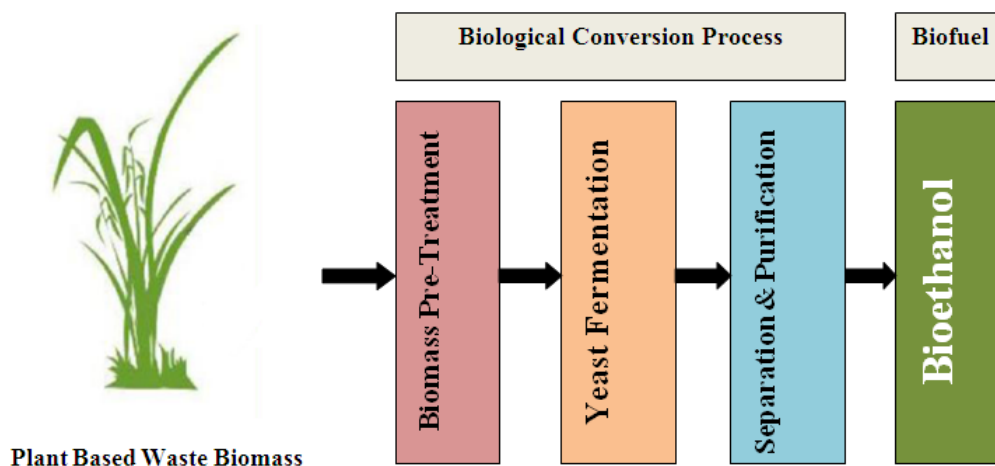


Figure-13

- **Anaerobic digestion (Figure-14)**

- ❖ AD is the conversion of organic material directly to a gas, termed biogas, a mixture of mainly methane and carbon dioxide with small quantities of other gases such as hydrogen sulphide.
- ❖ The biomass is converted by bacteria in an anaerobic environment, producing a gas with an energy content of about 20–40% of the lower heating value of the feedstock.

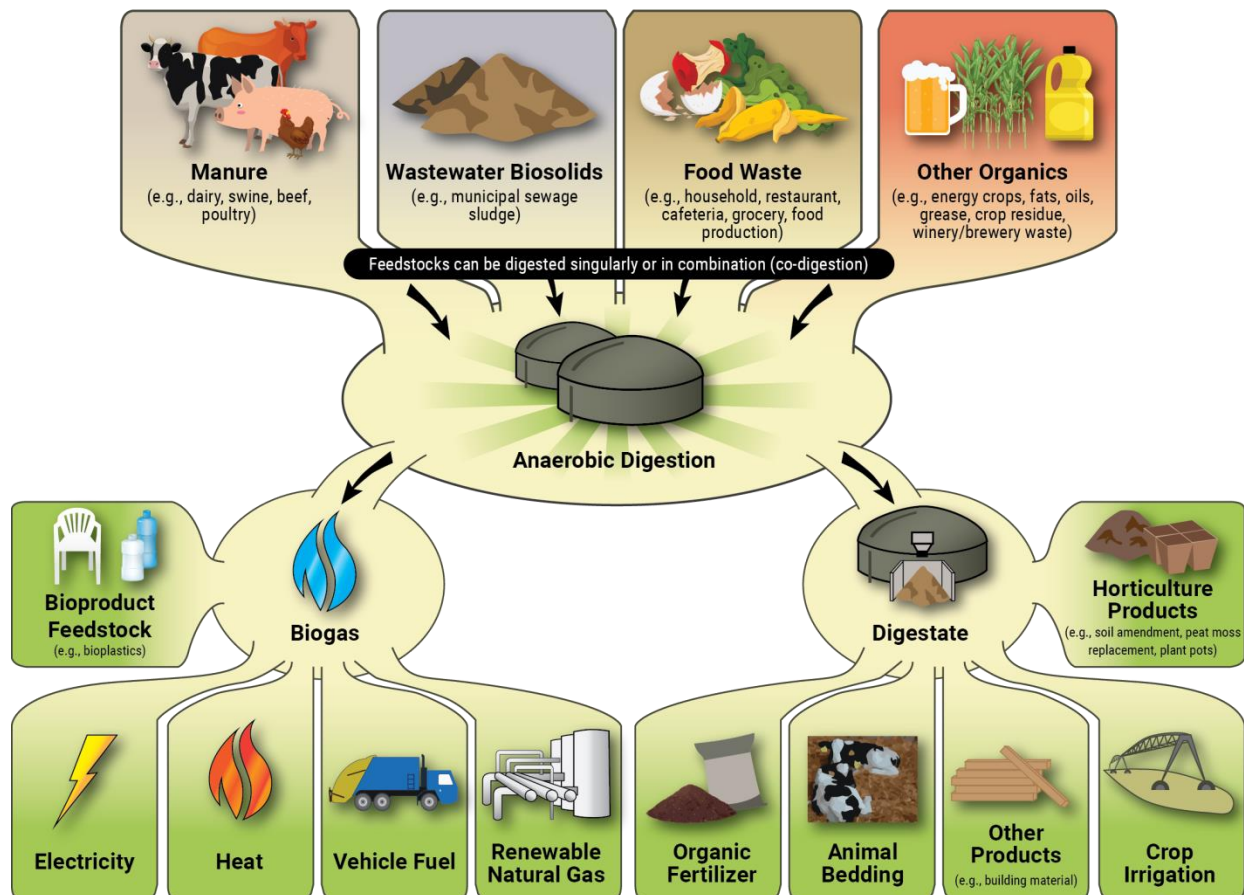


Figure-14



- **AEROBIC DIGESTION**

- ❖ Aerobic digestion is a process in sewage treatment designed to reduce the volume of sewage sludge and make it suitable for subsequent use.
- ❖ More recently, technology has been developed that allows the treatment and reduction of other organic waste, such as food, cardboard and horticultural waste.
- ❖ It is a bacterial process occurring in the presence of oxygen.
- ❖ Bacteria rapidly consume organic matter and convert it into carbon dioxide, water and a range of lower molecular weight organic compounds.

➤ **Mechanical extraction**

- Extraction is a mechanical operation used to produce oil from the seeds of various biomass crops, such as rapeseed, cotton and groundnuts.
- The process produces not only oil but also a residual solid or 'cake', which is suitable for animal fodder.
- Three tons of rapeseed are required per ton of rapeseed oil produced.
- Rapeseed oil can be processed further by reacting it with alcohol using a process termed esterification to obtain RME (Refined Methyl Ester) or bio-diesel.
- **Figure-7** depicts a generalised flow sheet for the production of methyl ester bio-diesel and the by-product glycerine.

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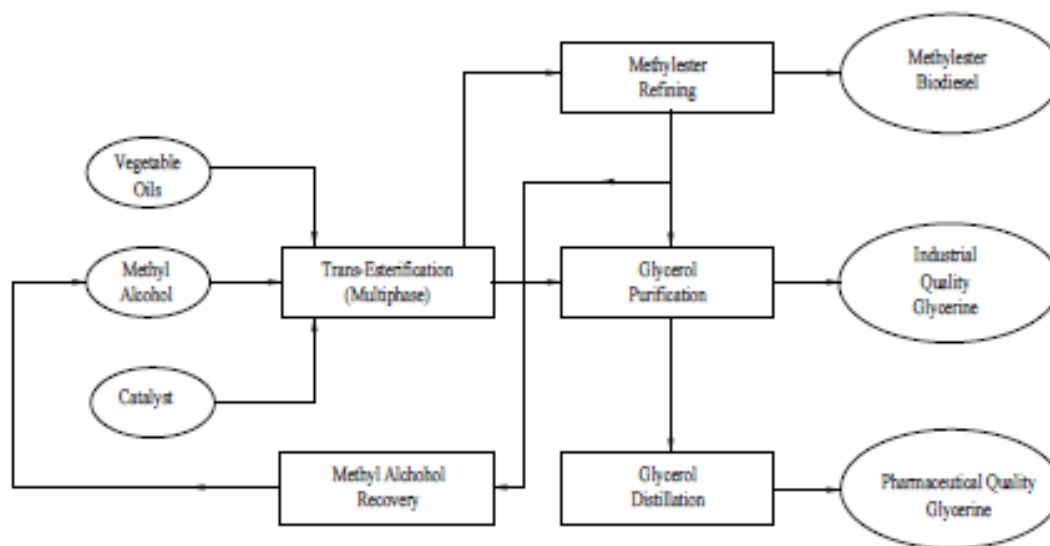


Figure-7: Flow sheet for the production of methylester bio-diesel and glycerin.

Table-1: Comparison between various Thermochemical Conversion Technologies

Process Routes	Advantages	Disadvantages
Gasification	• Very low emission. • Better control of process owing to lower operating temperature than combustion. • Produces small volume of gaseous products, as a result, less cleaning equipment is required. • Produces Char which can be used after activation.	• Capital cost is high. • Operation is complex because of the requirement of oxygen separation units. • Economy highly effected by costs associated with steam and oxygen. • A feed stock with higher ash content results in agglomeration problem.



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Combustion	• Mature and commercially viable technology. • The heat produced in the process can directly be used for power generation.	• Not environment friendly as increased emission problem. • No provision of storage of heat. • As volume of gaseous products is quite large, bigger gas cleaning equipments are required.
Hydrothermal Conversion (liquefaction)	• No pre-treatment of biomass required. • The amount of oxygen in bio-oil produced is lesser as compared to oil produced through pyrolysis.	• High pressure is required along with long residence time. • Expensive process due to cost of high pressure vessels and related equipment.
Pyrolysis	• Operating temperatures are lower compared to gasification and combustion. • Environmental friendly process as emissions are almost nil. • Produces variety of products including Pyro gas, bio-oil and bio char. • The solid product of bio char can be used after activation while other products can be potentially integrated in bio-refinery. • The transportation of products like bio-oil is easier than syn- gas or solid biomass.	• The bio-oil produced has been found to have lower heating value. • Products are usually immiscible with hydro carbons. • Due to corrosive nature of bio-oil, its long term storage is difficult. • The products produced need up gradation for application in turbines and engines.

TableII-Comparison between Biological and Thermo-chemical Conversions

Biological Conversion	Thermo-chemical Conversion
Uses microorganisms, enzymes with chemicals to utilize specific type of biomass.	Almost all types of biomass can be used as feedstock.
The productivity is low with high capital investment like large size reactor for specific conversion.	Due to complete chemical nature of the reaction , higher productivity is achieved.
Process is slow as carried out under ambient temperature condition.	The process is very fast.
Option for multiple products is limited due to specific nature of micro organism for a specific product.	Separation of products can be achieved easily using fractional separation train. Multiple products are obtained.
Secondary waste generation in the form of sludge poses another problem for its disposal.	In most of the cases biomass is completely used. No sludge or waste obtained.

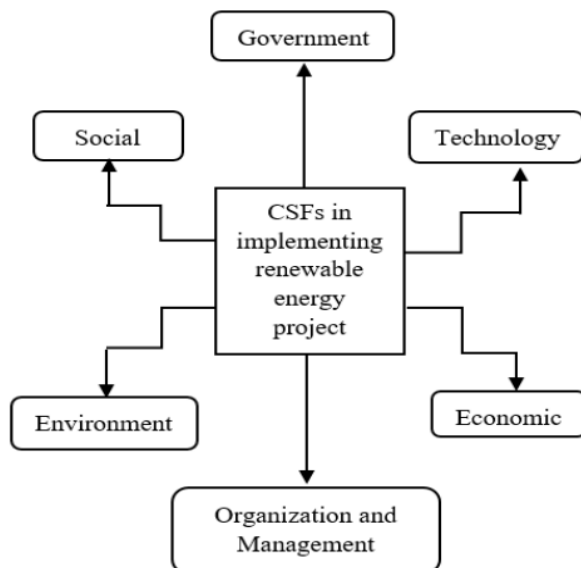


Figure-8: Critical success factor in implementing renewable energy projects.